
Algorithm 1 Convergence Test of a Series $\sum a_n$

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1: function TESTCONVERGENCE( $\sum a_n$ )
2:   if  $\left(\left|\frac{a_n}{b_n}\right| \nrightarrow \infty\right)$  then                                     ▷ (1) Comparison Test
3:     return TESTCONVERGENCE( $\sum b_n$ )
4:   else if  $(a_n \nrightarrow 0)$  then                                           ▷ (2) Divergence Test
5:     return "Divergent"
6:   else if  $\left(\lim_{n \rightarrow \infty} \left|\frac{a_{n+1}}{a_n}\right| < 1\right) \vee \left(\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1\right)$  then       ▷ (3) Ratio/Root Test
7:     return "Absolutely Convergent"
8:   else if  $\left(|a_n| = \frac{1}{n^p}\right) \wedge (p > 1)$  then                               ▷ (4)  $p$ -Series Test
9:     return "Absolutely Convergent"
10:  else if  $\left(\int_1^\infty |a_x| dx \nrightarrow \infty\right)$  then                               ▷ (5) Integral Test
11:    return "Absolutely Convergent"
12:  else if  $(a_n = (-1)^n |a_n|)$  then                                       ▷ (6) Alternating Series Test
13:    return "Conditionally Convergent"
14:  else                                                                    ▷ (7) Failure of Tests
15:    return "Divergent"

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Some explanatory paragraph before the columns.

Item 1.	3. Third item in a numbered list.	22. Second item in a numbered list.
Item 2.	4. Fourth item in a numbered list.	23. Third item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	5. Fifth item in a numbered list.	24. Fourth item in a numbered list.
Item 1.	6. Sixth item in a numbered list.	25. Fifth item in a numbered list.
Item 2.	7. Seventh item in a numbered list.	26. Sixth item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	8. Eighth item in a numbered list.	27. Seventh item in a numbered list.
Item 1.	9. Ninth item in a numbered list.	28. Eighth item in a numbered list.
Item 2.	10. Tenth item in a numbered list.	29. Ninth item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	11. First item in a numbered list.	30. Tenth item in a numbered list.
Item 1.	12. Second item in a numbered list.	31. First item in a numbered list.
Item 2.	13. Third item in a numbered list.	32. Second item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	14. Fourth item in a numbered list.	33. Third item in a numbered list.
Item 1.	15. Fifth item in a numbered list.	34. Fourth item in a numbered list.
Item 2.	16. Sixth item in a numbered list.	35. Fifth item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	17. Seventh item in a numbered list.	36. Sixth item in a numbered list.
1. First item in a numbered list.	18. Eighth item in a numbered list.	37. Seventh item in a numbered list.
2. Second item in a numbered list.	19. Ninth item in a numbered list.	38. Eighth item in a numbered list.
	20. Tenth item in a numbered list.	39. Ninth item in a numbered list.
	21. First item in a numbered list.	40. Tenth item in a numbered list.

Abstract

This document explains how generating functions use the concept of linear independence in the vector space of polynomials to count combinatorial objects. Key terms are defined and a concrete example is provided.

1 Key Terminology

- **Generating function:** A formal power series $G(x) = \sum_{n=0}^{\infty} a_n x^n$ encoding a sequence (a_n) .
- **Monomial:** A term x^n in a polynomial or power series.
- **Basis:** A set of vectors (here, monomials $\{1, x, x^2, \dots\}$) such that every element of the space is a unique linear combination of them.
- **Linear independence:** No nontrivial linear combination of basis elements equals zero; i.e., $\sum c_n x^n = 0$ implies all $c_n = 0$.
- **Coefficient:** The number multiplying x^n in a power series, often representing the count of objects of size n .

2 Core Relationship

Generating functions work because the monomials $\{1, x, x^2, \dots\}$ are linearly independent. Each coefficient a_n in $G(x)$ uniquely counts objects of size n . Linear independence ensures that collecting like terms (powers of x) is valid and unambiguous.

3 Example Problem

Find the number of non-negative integer solutions to $a + b + c = 10$ with:

- $a \in \{2, 3, 4\}$
- b even ($b \in \{0, 2, 4, \dots\}$)
- c odd ($c \in \{1, 3, 5, \dots\}$)

4 Encoding Constraints

Each variable's allowed values are encoded as a generating function:

$$\begin{aligned} A(x) &= x^2 + x^3 + x^4 \\ B(x) &= 1 + x^2 + x^4 + \dots = \frac{1}{1 - x^2} \\ C(x) &= x + x^3 + x^5 + \dots = \frac{x}{1 - x^2} \end{aligned}$$

5 Combining by Multiplication

The total generating function is $G(x) = A(x)B(x)C(x)$. Multiplying corresponds to summing exponents: selecting x^a, x^b, x^c gives x^{a+b+c} . The coefficient of x^{10} in $G(x)$ counts the solutions.

6 Why Linear Independence Matters

Because monomials are linearly independent, each power x^n in $G(x)$ has a unique coefficient. This guarantees that the coefficient of x^{10} counts only the desired combinations, with no overlap or ambiguity.

7 Solution

Expand:

$$\begin{aligned} G(x) &= (x^2 + x^3 + x^4) \cdot \frac{1}{1-x^2} \cdot \frac{x}{1-x^2} \\ &= \frac{x^3 + x^4 + x^5}{(1-x^2)^2} \end{aligned}$$

Expand the denominator:

$$\frac{1}{(1-x^2)^2} = \sum_{j=0}^{\infty} (j+1)x^{2j}$$

To get x^{10} , use:

- x^3 needs x^7 from denominator: not possible (only even powers).
- x^4 needs x^6 : $j = 3$, coefficient 4.
- x^5 needs x^5 : not possible.

Total: $0 + 4 + 0 = 4$.

Answer: There are 4 solutions.